Countable Additivity is Not First-Order Axiomatizable

Patrick S. Mahon

Abstract

In the first-order language of Boolean algebras equipped with a normalized finitely additive charge, no first-order theory characterizes those models whose charge extends to a σ-additive measure on the generated σ-algebra. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras. The proof is a single ultraproduct argument: Dirac charges on the finite-cofinite algebra are individually σ-additive, but their ultraproduct is the finite-cofinite charge, which fails σ-additivity — contradicting Łoś's theorem if any first-order axiomatization existed. The same obstruction shows that collective exhaustion — the necessary and sufficient condition for σ-additive extension in a directed system of Boolean algebras — cannot be derived from any first-order structural condition.

A property of structures is first-order axiomatizable in a language L if there exists a set of L-sentences whose models are exactly the structures satisfying it. Łoś's theorem gives the key diagnostic: if a property fails in an ultraproduct of structures that individually satisfy it, no first-order axiomatization exists.

Countable additivity distinguishes σ-additive measures from finitely additive charges. That it cannot be expressed by a single first-order sentence is clear: the condition quantifies over countable sequences, unavailable in first-order logic. Its non-first-order character is treated as background knowledge in the probability logic literature [4, 5, 3], but the sharper question — whether countable additivity admits a first-order *axiomatization* by any set of sentences, not merely a single formula — does not appear to have been settled by an explicit proof.

We settle it negatively.

Let $\mathcal{L}_{\mathrm{BA},\mu}$ be the one-sorted first-order language with:

- Boolean operations $\wedge, \vee, \neg, 0, 1$;
- for each rational $q \in [0,1] \cap \mathbb{Q}$, unary relation symbols $M_{\leq q}(x)$ and $M_{\geq q}(x)$, intended to mean $\mu(x) \leq q$ and $\mu(x) \geq q$ respectively.

A structure for $\mathcal{L}_{\mathrm{BA},\mu}$ is a Boolean algebra B together with a $[0,1]$-valued finitely additive probability μ, encoded through the truth of the predicates $M_{\leq q}$ and $M_{\geq q}$. Using rational comparison predicates rather than a function symbol into $[0,1]$ avoids introducing a second numeric sort.

Definition 1 (First-order theory of finitely additive probabilities)**.** Let T_{fa} be the $\mathcal{L}_{\mathrm{BA},\mu}$-theory axiomatizing:

(i) B is a Boolean algebra;

(ii) normalization: $\mu(0) = 0$ and $\mu(1) = 1$;

(iii) monotonicity: $x \leq y \Rightarrow \mu(x) \leq \mu(y)$;

(iv) finite additivity: for all rationals r, s with $r + s \leq 1$,

$$x \wedge y = 0 \wedge M_{\leq r}(x) \wedge M_{\leq s}(y) \;\Rightarrow\; M_{\leq r+s}(x \vee y),$$

and similarly for lower bounds.

The class of Boolean algebras with normalized finitely additive probability is first-order axiomatizable in $\mathcal{L}_{\mathrm{BA},\mu}$. Countable additivity, by contrast, requires quantification over an arbitrary countable sequence $(a_n)_{n\in\mathbb{N}}$ — not available in first-order logic — and so is not a sentence of $\mathcal{L}_{\mathrm{BA},\mu}$.

Proposition 2 (σ-additive extension is not first-order axiomatizable)**.** *There is no first-order $\mathcal{L}_{\mathrm{BA},\mu}$-theory $T \supseteq T_{\mathrm{fa}}$ such that, for every $(B,\mu) \models T_{\mathrm{fa}}$,*

$$(B,\mu) \models T \quad\iff\quad \mu \text{ extends to a } \sigma\text{-additive measure on } \sigma(B).$$

Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras.

Proof. Let E be the Boolean algebra of finite-cofinite subsets of $\mathbb{Q}$, fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$, and let δ_n be the Dirac charge at $q(n)$. Each (E,δ_n) is a model of T_{fa} in which δ_n is σ-additive. Suppose for contradiction that a theory T as described exists; then each $(E,\delta_n) \models T$.

Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter. By Łoś's theorem [6], the ultraproduct $\prod_{\mathcal{U}}(E,\delta_n)$ also models T. The charge induced on the diagonal copy of E is

$$\ell(A) = \lim_{\mathcal{U}} \delta_n(A) = \lim_{\mathcal{U}} \mathbf{1}_{q(n)\in A}.$$

If A is finite, $q(n) \in A$ for only finitely many n, so $\ell(A) = 0$. If A is cofinite, $q(n) \in A$ for cofinitely many n, so $\ell(A) = 1$ since $\mathcal{U}$ extends the cofinite filter. Thus ℓ is the finite-cofinite charge.

The sequence $A_k = \mathbb{Q} \setminus \{q(0),\ldots,q(k)\}$ satisfies $A_0 \supseteq A_1 \supseteq \cdots$ and $\bigcap_k A_k = \varnothing$, yet $\ell(A_k) = 1$ for all k. Hence the charge on the diagonal copy of E fails σ-additivity and does not extend to a σ-additive measure on $\sigma(E)$. Consequently the ultraproduct cannot satisfy any theory T characterizing σ-additive extension, contradicting Łoś's theorem applied to the assumption that each $(E,\delta_n) \models T$. □

Remark 3. The ultraproduct $\prod_{\mathcal{U}}(E,\delta_n)$ is an $\mathcal{L}_{\mathrm{BA},\mu}$-structure in its own right; ℓ is the charge induced on the diagonal copy of E inside it. Identifying the ultraproduct with (E,ℓ) is harmless for the contradiction, since the failure of σ-additivity is witnessed on the diagonal copy.

Let $(\mathcal{E}_i,\ell_i)_{i\in\iota}$ be a directed system of Boolean charge spaces: each $\mathcal{E}_i$ is a Boolean algebra and $\ell_i : \mathcal{E}_i \to [0,1]$ is a normalized finitely additive charge, with compatible connecting maps. *Collective exhaustion* (CE) is the following condition on the induced charge ℓ on the direct limit algebra $\mathcal{E} = \varinjlim \mathcal{E}_i$: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \varnothing$, we have $\ell(E_n) \to 0$. It is the necessary and sufficient condition under which a compatible finitely additive family extends to a σ-additive measure on the σ-algebra generated by the system.

Corollary 4 (CE is not first-order)**.** *In the first-order language $\mathcal{L}_{\mathrm{BA},\mu}$, no first-order theory characterizes those models satisfying CE. Equivalently, no accumulation of first-order conditions on the Boolean-algebraic structure can imply collective exhaustion.*

The finite-cofinite charge is the canonical witness: it satisfies every first-order structural condition yet assigns persistent unit mass to the decreasing sequence $A_k = \mathbb{Q} \setminus \{q(0),\ldots,q(k)\}$, whose intersection is empty. CE fails precisely because the mass is not exhausted.

Foundational disputes about probability are not terminological. They concern which extra structure licenses the passage from finite coherence to probability, and Corollary 4 shows that some such structure is logically unavoidable: no first-order condition can close the gap. For a full development of CE as the admissibility condition for σ-additive extension in a directed observational system, see [7].

The proof produces one purely finitely additive charge as an ultraproduct of σ-additive ones; that single witness suffices for the non-axiomatizability conclusion. A stronger question is which purely finitely additive charges on a Boolean algebra arise as pointwise ultralimits of σ-additive probabilities on the same algebra. On a σ-algebra the answer is none: the identity $\mu_n(A) = \sum_k \mu_n(A_k)$ holds exactly for each n, so any pointwise ultralimit of σ-additive probabilities is itself σ-additive [2], IV.9.8. On a general Boolean algebra the question is open: ultrafilter-generated $\{0,1\}$-valued charges are representable (the present Dirac construction gives this), but whether the same holds for all purely finitely additive charges — and whether the answer is set-theoretically sensitive — is not known [8, 1].

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